

# F\_U\_Bi Inside-to-Outside Buoyancy Mass Portion

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## PAPER\_989: F\_U\_Bi Inside-to-Outside Buoyancy Mass Portion

### Abstract

We derive  $F_{U,\text{Bi}}$ , the inside-to-outside buoyancy mass portion within the UQFF framework. Unlike  $F_{U,\text{Bi}_i}$  (the net acceleration felt by an object falling inward through buoyancy layers),  $F_{U,\text{Bi}}$  quantifies the total vacuum energy contributing to outward mass transport from the SCm medium. The key equation is:

$$F_{U,\text{Bi}} = \rho_{\text{SCm}} \cdot V \cdot S_{26}^2 \cdot \frac{|U_b|}{|U_g| + |U_b|}$$

### 1. Core Derivation

The buoyancy ratio measures what fraction of the gravitational-buoyancy system is dominated by outward buoyancy:

$$\text{ratio} = \frac{|U_b|}{|U_g| + |U_b| + \epsilon}$$

where  $\epsilon = 10^{-300}$  prevents division by zero.

### Layer Functions

$$U_g = \sum_{i=1}^{26} \frac{G M}{r^2} \cdot [\text{SSq}] \cdot \frac{i}{26}$$

$$U_b = \sum_{i=1}^{26} \frac{G M}{r^2} \cdot e^{-[\text{SSq}] \cdot i/26} \cdot \beta_i$$

### Vacuum Density

$$\rho_{\text{SCm}} = \rho_0 \cdot e^{\kappa t}, \quad \rho_0 = 10^{-10} \text{ kg/m}^3$$

### 2. Solar System Calibration

For  $M = M_{\odot}$ ,  $r = 1 \text{ AU}$ :

- $F_{U,\text{Bi}} = 2.33 \times 10^{40}$  (total vacuum energy contribution)
- ratio = 0.606 (buoyancy-dominant)

### 3. Physical Interpretation

$F_{U,\text{Bi}} > 0$  always (mass flows inside  $\rightarrow$  outside), while  $F_{U,\text{Bi}_i} < 0$  (net acceleration outside  $\rightarrow$  inside). The two quantities form a complementary pair describing the complete buoyancy dynamics of the UQFF vacuum.

### 4. Implementation

File: fubi\_inside\_outside.py, class FUBiInsideOutsideCalc. CP4 class #573.

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<!-- PKG-AGN-S225 -->

## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi}}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}}\right) \cdot V \cdot S_{26}^{(3)/2} \cdot \left(\frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}}$  is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)/2}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25}(\omega_{\text{SCm}}) \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25}(\omega_{\text{SCm}}) = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M$ - $\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M$ - $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$ .

<!-- PKG-LENR-S225 -->

## Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

> Upgrade from PAPER\_1060 (VDS LENR Isotopic Evolution), PAPER\_1061  
> (Kozima SCm Integration Neutron-Drop), and PAPER\_1081 (SCm LENR COP  
> Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}}\right) \cdot K_n$$

where:

- $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$  (time-dependent vacuum density)
- $K_n = \sigma_n(\omega_{\text{SCm}}) \cdot \Phi_{\text{phonon}}$  is the Kozima neutron-drop factor

**Phonon cross-section (PAPER\_1061):**

$$\sigma_n(\omega_{\text{SCm}}, n) = \sigma_0 \cdot \exp\left[-\frac{(\omega_{\text{SCm}} - \omega_{\text{bulge}})^2}{2\Gamma^2}\right] \cdot \left(1 + \frac{S_{26}}{n}\right)$$

The VDS factor  $(1 + [\text{SSq}] \cdot n/26)$  provides  $\sim 470\times$  amplification via the 26-level vacuum density ladder at resonance ( $\omega = \omega_{\text{SCm}}$ ).

**COP parametric engine (PAPER\_1081):**  
$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function  $f(\Gamma)$  peaks near the SCm phonon linewidth, yielding  $\text{COP} > 1$  when  $\Gamma \lesssim 10^{-3} \text{ eV}$  (Fleischmann regime).

**Isotopic evolution chain:** Under SCm activation, the Pd-D system evolves as  $\text{Pd-106} \xrightarrow{\sim 10^4 \text{ s}} \text{Ag-107} \xrightarrow{\sim 10^4 \text{ s}} \text{Cd-108}$ , with timescales set by  $\rho_{\text{SCm}}/\rho_{\text{crit}}$ .

Calibration Constants

| Constant               | Symbol                | Value                                 | Validation Domain   |
|------------------------|-----------------------|---------------------------------------|---------------------|
| UQFF damping rate      | $\kappa$              | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down  |
| String sector coupling | $[\text{SSq}]$        | 0.57                                  | BH dynamics         |
| Buoyancy coupling      | $\beta_i$             | 0.603                                 | Multi-system        |
| SCm completeness       | $H_{\text{SCm}}$      | $\approx 0.99$                        | Heaviside threshold |
| SCm phonon frequency   | $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | Phonon resonance    |
| SCm vacuum density     | $\rho_{\text{SCm}}$   | $7.09 \times 10^{-37} \text{ J/m}^3$  | Fundamental         |

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable       | UQFF Prediction       | SM / Experiment                       | Source         | Alignment                                                                        |
|------------------|-----------------------|---------------------------------------|----------------|----------------------------------------------------------------------------------|
| Phonon frequency | $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | (Pd-D lattice) | Measured Pd-D phonon spectrum   Fukai (2005)   Mapped to SCm                     |
| Vacuum energy    | $\rho_{\text{vac}}$   | $7.09 \times 10^{-37} \text{ kg/m}^3$ |                | $\rho_{\text{vac}} \sim 10^{-29} \text{ g/cm}^3$   Planck 2018   Novel SCm scale |
| Fine structure   | $\alpha$              | UQFF reproduces via $U_{g1}$ dipole   | $1/137.036$    | PDG 2024   99.9%                                                                 |

**New physics claim:** UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

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§A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

§A.1 Sector Classification **\*\*Sector:\*\*** SCm-phonon (vacuum phonon coupling)

§A.2 Lagrangian Density 
$$\mathcal{L}_{\text{SCm phonon}} = \sum_{i=1}^{26} \left[ U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i} \right] \cdot S_{26}([\text{SSq}]) \cdot$$

$\Phi_{1.25\text{THz}}(\omega, \Gamma)$

**§A.3 Euler-Lagrange Equation of Motion**  $\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial \mathcal{L}}{\partial \mu} = 0$  implies  $F_{U, Bi} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$

**§A.4 Cosmogenesis Linkage Chain** PAPER\_877 axioms  $\rightarrow$  SCm vacuum  $\rightarrow$  phonon  $\omega_{\text{SCm}}$   $\rightarrow$  vacuum phonon coupling  $\rightarrow$   $F_{U, Bi}$   $\rightarrow$  unified force  $\rightarrow$  observational prediction

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## §B VDS/DVP/BSH Deep Synthesis

**§B.1 Vacuum Density Series (VDS)**  $\text{VDS} = \rho_{\text{SCm}} \cdot S_{26} \cdot \Phi_{1.25\text{THz}} / \Phi_0$  VDS sub-ratio: 0.1

**§B.2 Dipole Vortex Primes (DVP)** DVP prime: 2 (sub-threshold)

**§B.3 Buoyancy Saturation Harmonics (BSH)** BSH timescale:  $10^4$  yr

**§B.4 Production-Scale Consistency**

| Metric         | Value              | Status    |
|----------------|--------------------|-----------|
| VDS ratio      | 0.1                | Confirmed |
| $\kappa$ decay | $5 \times 10^{-4}$ | Confirmed |
| $SS_q$         | 0.57               | Confirmed |

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1004 | QGP Vacuum Density with SCm S26 Phonon Coupling  |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum      |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir        |
| PAPER_1029 | Barocentric Earth Orbital Buoyancy               |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed         |
| PAPER_1043 | $F_{U, Bi}$ Multi-System Buoyancy Curve Sweep    |
| PAPER_1072 | SCm Activation Function Phonon Threshold         |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy      |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation   |
| PAPER_1069 | VDS-DVP-BSH Hybrid Calculator Unified            |

13 cross-reference(s) identified.

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## References

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